DPP	DEPARTMENT POLICY AND PROCEDURE		
Title	**INTUBATION AND EXTUBATION IN PICU**		
Department	**Pediatric Intensive Care Unit**		
Applies to	**All PICU Medical and Nursing Staff**		
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1.0 PURPOSE

1.1. To provide patent airway.

1.2. To provide route for short term mechanical ventilation.

1.3. To facilitate removal of pulmonary secretions.

1.4. To relieve CO2 retention in clients with chronic pulmonary disease.

1.5. To treat acute respiratory failure.

1.6. To describe process of securing a proper elective endotracheal intubation of critically ill children in PICU setting.

2.0 DEFINITION/ABBREVIATION

2.1. **ETT** - Endotracheal tube

2.2. **Endotracheal intubation** - is an insertion of flexible tube through the mouth or nose into the trachea beyond the vocal cords that acts as an artificial airway. Endotracheal intubation is a medical procedure in which a tube is placed into the windpipe (trachea) through the mouth or nose.

3.0 RESPONSIBILITY

3.1 **Physician** is responsible in assessing patient's condition before and after endotracheal tube insertion and writing an order for any procedure and medication needed.

3.2 **Head Nurse** is responsible to make sure that the unit staffs are well knowledge in assisting insertion of endotracheal tube and all needed equipment are available in the unit for emergency cases.

3.3 **Charge Nurses** are responsible to delegate the procedure to a competent nurse before the procedure and to double check physicians' orders if carried out and help the staff nurse throughout the procedure including double checking medications.

3.4 **Staff nurses** are responsible in carrying out physician's orders, preparing needed equipment before the procedure and assisting physician. It is their responsibility as well to frequently reassess patient condition.

4.0 POLICY

4.1. Procedure should be performed by an experienced physician.

4.2. A physician order is required.

4.3. Sterile technique must be observed throughout the procedure.

4.4. Endotracheal intubation should be planned process in PICU setting. Anticipation may prevent respiratory arrest. Personnel and equipment prepration will facilitate smooth and endotracheal intubation and avoid complication.

4.5. The primary goal of endotracheal intubation is secure the airway and provide adequate oxygenation and ventilation. Endotracheal intubation should be performed via the oral route initially, since this is the easiest in children. However, changing to nasal can be done in later stage after patient is more stable.

4.6. Endotracheal intubation must be done by the Intensive Care Unit doctor and to be assisted by a qualified nursing personnel who has adequate knowledge on the procedure.

4.7. All personnel should have BLS and PALS certificates.

4.8. If difficulties anticipated, anesthesia department consultation should be done.

4.9. In case of unexpected respiratory arrest i.e. emergency situation; endotracheal intubation should be performed ASAP (as soon as possible) by competent personnel.

4.10. There should be a nurse, respiratory therapist and ICU physician at the bedside.

5.0 PROCEDURE (S)

5.1. **Indication for elective endotracheal intubation must be observed.**

5.1.1. For airway patency

5.1.2. Apnea

5.1.3. Requirement for positive pressure ventilation due to hypoxia and/or hypercarbia

5.1.4. Significant cardiovascular compromise or shock

5.1.5. Neurological: seizure, weakness, and traumatic brain injury within adequate airway protection.

5.2. Critically ill child should be connected to PICU monitor with alarm on and preferable with at least one function IV access. Do not delay resuscitation and intubation until IV access is established.

5.3. Standard precautions must be observed.

5.4. If possible, the patient is allowed to breath spontaneously while receiving 100% oxygen for at least 3 minutes.

5.5. Position the patient allowing easy access to the airway.

5.6. Administer appropriate medications according to the clinical condition.

5.7. Monitor vital signs while intubating the patient.

5.8. Confirm the endotracheal tube site by CO2 detector, auscultate for the breath sounds over lateral chest wall and stomach, and check for bilateral chest expansion.

5.9. Secure endotracheal tube noting position and presence of leak around ETT.

5.10. Verify breath sounds again after taping the tube.

5.11. Chest x-ray post intubation.

5.12. Blood gas 30 minutes' post intubation.

5.13. Initiate appropriate sedation

5.14. Oral care every 4 hours and PRN.

5.15. Verify ETT position and securely tape it every day.

5.16. Check tube placement every 8 hours and record the level in cm at gum line, teeth or inner nares for nasally intubated patients.

5.17. Auscultate for breath sounds every 4 hours and record it in respiratory flow sheet.

5.18. If tube is cuffed, ensure minimal cuff leak is present every 8 hours.

5.19. **CRITERIA FOR EXTUBATION**

5.19.1. Extubation is appropriate when the indications for intubation are no longer present. In general, this means that the work of breathing has decreased to a level manageable by the patient. There is no single clinical or laboratory criterion which will indicate the mnemonic readiness for extubation. A review of systems using the mnemonic CALMS (CNS, Airway, Lungs, Muscles, and System status) is often helpful.

5.19.2. The child should have an intact gag and sough as well as a central drive to breath.

5.19.3. If serious airway obstruction is anticipated, the patient should be extubated in OR.

5.19.4. The child should not exceed more than 40-50% O2, on CPAP or a low rate, the child should not exhibit hypercarbia, excessive tachypnea or dypsnea. A chest x-ray or knowledge as to the nature and quantity of secretions before extubation is helpful.

5.19.5. The muscular strength of a sick patient is often diminished. Can the patient cough, take deep breaths, breath spontaneously?

5.19.6. A negative inspiratory force (NIF) and forced vital capacity occasionally helpful in assessing muscle strength. However, observation of CPAP is the most useful test.

5.19.7. The child systemic condition i.e. hemodynamic status acid-base balance. Intracranial pathology, etc. should all be stable and manageable without an endotracheal tube before extubation is considered.

5.19.8. If the patient fulfils the extubation criteria in terms of clinical. Laboratory, and x-ray definition as established by the physician for that specific child, then preparations are made for the procedure and for the possibility of emergent reintubation if the child fails.

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5.19.9. If the PICU medical staff on service is inexperienced at intubation, an anaesthetist must be present. In all cases should be guided by the anticipated risk of decompensation after extubation

5.19.10. Technique

5.19.10.1. The endotracheal tube is suctioned after which the child is placed on 100% O2 via the bag

5.19.10.2. The stomach should be emptied via NG tube (patient should have been NPO for 4-6 hours before extubation)

5.19.10.3. After a full inspiration, the trachea is extubated under positive pressure.

5.19.10.4. Supplemental oxygen is provided by mask and the child suctioned as needed.

5.19.10.5. A post-extubation blood gas and a chest x-ray are done as indicated.

5.19.10.6. A procedure note is documented as adequacy of respiratory function and interventions carried out.

6.0 EQUIPMENTS

6.1. Proper size ambo bag valve mask connected to 100% oxygen

6.2. Proper size oropharyngeal/ nasopharyngeal airway.

6.3. Proper size suction catheter and yankuer

6.4. Laryngoscope and 2 blades (starght and curved) of appropriate size and with good bright light.

6.5. styled and magill forceps

6.6. endotracheal tube: Size (4+age in years). One up and one down from anticipated size needed.

6.7. Cuffed endotracheal tubes if child 2-8 years or clinically indicated

6.8. Three ml syringe to inflate the cuff in case of cuff tube is used

6.9. CO2 detector (Yellow=Yes)

6.10. Stethoscope

6.11. Nasogastric tube in situ

6.12. Tincture of benzoin

6.13. Adhesive tape (water proof tape)

6.14. Scissors

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6.15. Sterile normal saline solution

6.16. Adapter to fit ventilator

6.17. Ventilator with proper initial setting

6.18. Yankuer suction

6.19. Cool aerosol mask

7.0 ATTACHMENT/FORMS

7.1. N/A

8.0 REFERENCES

8.1. King Fahad Hospital, Policy and Procedure for Intubation and Extubation in PICU, Policy Number IEP-PICU-2/3 (5)

9.0 REVISION CLAUSE

9.1. N/A

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